CHAPTER C

Assessments and annual preparation

Versions T1 and T2 are available to pupils. Versions A and B and all the solutions are reserved for the teacher. The papers are original; they are not presented as official past papers. The approaches change between versions: direct or inverse model, optimisation or relative position, direct or reversed tree, projection or locus.

EV1-T1 · Common written test

Practice paper . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 9 C. 8 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 60 B. 64 C. 96 D. 100

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. -6

8. The slope between (0; 3) and (2; 9) is
   A. 2 B. -3 C. 6 D. 3

9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 7 B. 5 C. 12 D. -1

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A book collection (5 points)

A library has 120 books at the end of the reference month, called month 0. At the end of each following month, it adds 18 books to its collection; no books are removed. Let u_n the number of books available at the end of month n, for every natural number n.

1. Determine the type of (u_n), its common difference or ratio and its expression in terms of n. (1 pt)

2. Calculate u_8, then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not count only distinct books. (2 pts)

3. Determine the first month in which the collection strictly exceeds 300 books. Check the previous month. (2 pts)

Exercise 2 · Area and shape choice (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures x metres, the other measures $14 - x$ metres. Its area is modelled by the function A defined on [0; 14] by $A(x) = x(14 - x)$, in square metres. The endpoints represent flattened rectangles.

1. Expand, then write A in vertex form. (1 pt)

2. Determine the maximum area and the corresponding value of x without using derivatives. (2 pts)

3. Solve $A(x) \geq 40$ on the domain. (2 pts)

Exercise 3 · A right angle (4 points)

The plane is equipped with an orthonormal coordinate system. Consider the points $A(0; 1)$, $B(3; 2)$ and $C(-1; 4)$. The lengths are expressed in the unit of the coordinate system and areas in square units.

1. Calculate the coordinates of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)

2. Prove that these vectors are orthogonal and state the type of triangle. (1 pt)

3. Calculate the length BC then the area of triangle ABC. (2 pts)

Solutions EV1-T1

MCQ: 1 B, 2 C, 3 A, 4 C, 5 D, 6 D, 7 B, 8 D, 9 D, 10 B, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · A book collection

1. $u_n = 120 + 18n$, arithmetic with common difference 18. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. u_8 = 264; nine terms, sum $\dfrac{9(120+264)}{2}$ = 1728. This sum adds up monthly totals, not distinct books. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $120 + 18n > 300 \iff n > 10$. First index 11; $u_{10} = 300$, $u_{11} = 318$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Area and shape choice

1. $A = -x^2 + 14x = 49 - (x-7)^2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $A \leq 49$, equality at 7. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $49 - (x-7)^2 \geq 40 \iff (x-7)^2 \leq 9$, hence [4; 10]. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · A right angle

1. $\overrightarrow{AB} = (3;1)$, $\overrightarrow{AC} = (-1;3)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Product $-3 + 3 = 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. $BC = \sqrt{(-4)^2 + 2^2} = 2\sqrt{5}$; perpendicular sides of length $\sqrt{10}$, area 5. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV1-T2 · Common written test

Practice paper . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $4 + 2 \times 3$.
   A. 18 B. 9 C. 13 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 100 B. 96 C. 64 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$

7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 3 B. -3 C. 6 D. 2

9. The mean of $4, 6, 8$ is
   A. 18 B. 6 C. 7 D. 4

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 6 B. 16 C. 8 D. 0

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Two growth processes (5 points)

We compare two models for a workshop's output. On day 0, each predicts 30 objects. Model A predicts 10 extra objects each day; model B predicts a doubling daily. The outputs on day n are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. A cumulative total here means the sum of the outputs on the given days.

1. State the type of each sequence and calculate the production on day 2. (1 pt)

2. Calculate the cumulative production for days 0 to 3 in each model. (2 pts)

3. Check that model B exceeds model A on day 1, then explain why this gap remains positive on every following day by comparing the daily increases. (2 pts)

Exercise 2 · Imposed roots (5 points)

We are looking for a polynomial function f of degree 2, defined on $\mathbb{R}$. Its graph crosses the x-axis at the x-values -1 and 5, and it passes through the point (0; 10). This information should be enough to determine the function.

1. Give its factorised form. (1 pt)

2. Determine its sign, then solve $f(x) \leq 0$. (2 pts)

3. Give its vertex form and its maximum. (2 pts)

Exercise 3 · A controlled approximation (4 points)

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We want to approximate $f(3{,}02)$ using the tangent to its curve at the point with x-coordinate 3. Calculations are done without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)

2. Deduce $f'(3)$, then the equation of the tangent at the point with x-coordinate 3. (1 pt)

3. Use the tangent to approximate $f(3{,}02)$. Then calculate the error "exact value minus approximation". (2 pts)

Solutions EV1-T2

MCQ: 1 D, 2 C, 3 B, 4 D, 5 A, 6 A, 7 C, 8 A, 9 B, 10 C, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Two growth processes

1. A arithmetic, B geometric; $a_2 = 50, b_2 = 120$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. A : $\dfrac{4(30+60)}{2} = 180$; B : $30(1+2+4+8) = 450$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $b_1 = 60 > a_1$ = 40. For $n \geq 1$, increase in B 30 $\times 2^n \geq 60$, greater than 10; each step increases the gap. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Imposed roots

1. $f = a(x+1)(x-5)$; $-5a$ = 10, therefore $a = -2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. Negative leading coefficient; f positive between the roots, negative elsewhere. Solutions] $-\infty; -1] \cup [5; +\infty[$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $f = -2x^2 + 8x + 10 = -2(x-2)^2$ + 18. Maximum 18 at 2. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · A controlled approximation

1. Rate 6 + h. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. $f\prime(3) = 6,\ f(3) = 10$; $y = 6x - 8$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*

3. Approximation 10.12; exact 10 ,1204. Exact minus approximate 0 $,0004 = h^2$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV1-A · Common written test

Paper reserved for the teacher . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 10 D. 21

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 1 C. 10 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 100 B. 60 C. 64 D. 96

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of $2x + 5 = 0$ is
   A. 10 B. $\frac{-5}{2}$ C. -10 D. $\frac{5}{2}$

8. The slope between (0; 5) and (2; 11) is
   A. 6 B. 3 C. 2 D. -3

9. The mean of 5, 7, 9 is
   A. 8 B. 21 C. 5 D. 7

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 9 B. 7 C. 1 D. 20

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Decreasing reserve (5 points)

A reservoir initially contains 192 L of water. At each step, half of the water still present, without adding water. We denote u_n be the volume, in litres, remaining after n steps; index 0 corresponds to the initial state.

1. Define the sequence by its first term and its common difference or ratio. (1 pt)

2. Calculate the volumes for indices 0 to 4 and their sum. (2 pts)

3. Determine the first index at which the volume is strictly less than 10 L; describe a threshold loop. (2 pts)

Exercise 2 · Parameter and tangent (5 points)

For a given real m , define on $\mathbb{R}$ the function $f_m(x) = x^2 - 2mx + 4$. The number m is a parameter. The first two questions relate to all its values; the last concerns only $m = 1$.

1. Calculate the discriminant of the polynomial f_m as a function of m. (1 pt)

2. Determine, depending on m, the number of real solutions of $f_m(x) = 0$. (2 pts)

3. We now fix m = 1. Calculate the rate of change between 2 and 2 + h, for $h \neq 0$, then the equation of the tangent at the point with x-coordinate 2. (2 pts)

Exercise 3 · Locus via the midpoint (4 points)

Two fixed points A and B of the plane are 8 units apart. Let I be the midpoint of segment [AB] and M a variable point of the plane. We seek to describe sets of points using the scalar product $MA \cdot MB$.

1. Express $\overrightarrow{MA} \cdot \overrightarrow{MB}$ in terms of MI, justifying the identity used. (1 pt)

2. Describe the set of points M for which this product equals 9. (1 pt)

3. Describe in turn the sets obtained when the product equals -16, then -20. (2 pts)

Solutions EV1-A

MCQ: 1 B, 2 A, 3 D, 4 B, 5 A, 6 B, 7 B, 8 B, 9 D, 10 C, 11 A, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Decreasing reserve

1. $u_0 = 192$, $u_{n+1} = \dfrac{u_n}{2}$; ratio $\dfrac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. $192, 96, 48, 24, 12$, sum 372. *Marking: 2 points. Method or justification: 1; calculation and conclusion : 1.*

3. $u_4 = 12$, u_5 = 6: index 5. Initialise n=0, u=192; while u is at least 10, divide u by 2 and increase n. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Parameter and tangent

1. $\Delta = 4m^2 - 16 = 4(m^2 - 4)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Two roots if $m < -2$ or $m > 2$; one if $m = \pm 2$; none if $-2 < m < 2$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. For m=1, rate 2 + h, derivative 2 and $f(2)$ = 4; tangent $y = 2x$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Locus via the midpoint

1. $MI^2 - 16$, by the scalar product identities. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. MI^2 = 25 : circle with centre I, radius 5. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

3. For -16, MI^2 = 0 : I alone. For -20, $MI^2 = -4$: empty. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV1-B · Common written test

Paper reserved for the teacher . Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $6 + 2 \times 3$.
   A. 11 B. 12 C. 15 D. 24

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{5}{3}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 11 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 70 C. 10 D. 40

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 96 C. 60 D. 100

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{21}{20}$ B. $\frac{4}{5}$ C. 1 D. $\frac{9}{10}$

7. The solution of $2x + 6 = 0$ is
   A. 3 B. 12 C. -3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 6 D. 2

9. The mean of 6, 8, 10 is
   A. 8 B. 24 C. 6 D. 9

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 8 B. 24 C. 10 D. 2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Growth by pattern (5 points)

A family of patterns is indexed by the integers $n \geq 1$. The number of dots in the pattern of index n is given by $u_n = n^2 + 2n$. In particular, we seek the first pattern with at least 99 dots; no figure is required to use the given formula.

1. Calculate u_1, u_2, u_3. (1 pt)

2. Calculate $u_{n+1} - u_n$ then determine how it varies. (2 pts)

3. From which index does the pattern contain at least 99 dots? Justify without an endless search. (2 pts)

Exercise 2 · Two dimensions and one constraint (5 points)

A rectangle has a perimeter of 26 m and an area of 40 square metres. Let x and y be its two dimensions, expressed in metres. These two numbers are strictly positive. We will then compare this rectangle with the other rectangles having the same perimeter.

1. Write the relationships between the two lengths x and y. (1 pt)

2. Find the dimensions using a quadratic equation. (2 pts)

3. Among the rectangles with the same perimeter, determine the maximum area and compare it with 40. (2 pts)

Exercise 3 · Angle and then approximation (4 points)

Questions 1 and 2 concern two vectors $\vec{u}$ and $\vec{v}$ with magnitudes 2 and 3 respectively, such that $\vec{u} \cdot \vec{v} = 3$. Question 3 is independent: it concerns approximation by a tangent to the square function.

1. Find the measure in degrees of the unsigned angle between $\vec{u}$ and $\vec{v}$. (1 pt)

2. Calculate the exact value of $\|\vec{u} - \vec{v}\|$. (1 pt)

3. Independently, the tangent to the square function at 1 has equation $y = 2x - 1$. Use it to approximate $1{,}03^2$, then calculate the difference between the exact value and the approximation. (2 pts)

Solutions EV1-B

MCQ: 1 B, 2 B, 3 D, 4 A, 5 D, 6 C, 7 C, 8 B, 9 A, 10 C, 11 B, 12 C. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Growth by pattern

1. $3, 8, 15$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Difference 2 $n + 3 > 0$: strictly increasing. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $n^2 + 2n - 99 = (n - 9)(n + 11) \geq 0$ with $n \geq 1$: $n \geq 9$; $u_9 = 99$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Two dimensions and one constraint

1. $x + y = 13$, xy = 40, x,y positive. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Roots of $t^2 - 13t + 40 = (t - 5)(t - 8)$: 5 m and 8 m. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*

3. $A(x) = x(13 - x) = \frac{169}{4} - (x - \frac{13}{2})^2$, maximum $\frac{169}{4} = 42{,}25$ m^2. Gap $\frac{9}{4}$ m^2. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Angle then approximation

1. $\cos\theta = \frac{3}{6} = \frac{1}{2}$, therefore 60°. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Square 4 + 9 − 6 = 7; norm $\sqrt{7}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

3. Approximation 1.06; exact 1.0609; error 0.0009. This independent question checks local differentiation. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV2-T1 · Mock exam 1

Practice paper . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14·16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $2 + 2 \times 3$.
   A. 8 B. 12 C. 11 D. 7

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 100 C. 64 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$

7. The solution of $2x + 2 = 0$ is
   A. -4 B. -1 C. 1 D. 4

8. The slope between (0; 2) and (2; 8) is
   A. 6 B. 2 C. 3 D. -3

9. The mean of $2, 4, 6$ is
   A. 4 B. 2 C. 12 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 4 B. 8 C. 6 D. -2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Manufacturing and inspection (5 points)

Two workshops A and B produce 30 per cent and 70 per cent respectively of the items made by a company. Of the items from A, 10 per cent are defective; the proportion is 20 per cent for B. An item is chosen at random from the total production. The events A and D mean "comes from workshop A" and "is defective" respectively.

1. Construct a weighted tree whose first level distinguishes the two workshops. (1 pt)

2. Calculate $P(D)$, then the probability that a defective item comes from A. (2 pts)

3. Are the events A and D independent? Justify numerically. (2 pts)

Exercise 2 · Minimum of a quotient (5 points)

The function f is defined on the interval $]0; +\infty[$ by $f(x) = x + \dfrac{9}{x}$. We study its variations and certain tangents to its curve. The requested coordinates must be given in exact form.

1. Calculate and factorise the derivative. (1 pt)

2. Study the variation and give the minimum. (2 pts)

3. Determine the points where the tangent has slope $\dfrac{3}{4}$. (2 pts)

Exercise 3 · Projection and circle (4 points)

In an orthonormal coordinate system, consider the point $P(5; 2)$ and the line d with equation $x + y - 3 = 0$. Let H be the orthogonal projection of P on d.

1. Give a non-zero normal vector to d. (1 pt)

2. Determine the coordinates of H and check the membership and orthogonality conditions. (1 pt)

3. Give an equation of the circle with centre P through H. (2 pts)

Solutions EV2-T1

MCQ: 1 A, 2 D, 3 D, 4 B, 5 B, 6 C, 7 B, 8 C, 9 A, 10 C, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Manufacturing and inspection

1. First level: 0 ,3 and 0,7. After A: 0 ,1 and 0,9; after B: 0 ,2 and 0,8. Each pair sums à 1. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $P(D) = 0{,}03 + 0{,}14 = 0{,}17$; $P_D(A) = \dfrac{3}{17}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. No: $P_A(D) = 0{,}1 \neq 0{,}17$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Minimum of a quotient

1. $f\prime = \dfrac{(x-3)(x+3)}{x^2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Decreases down to 3 then increases; minimum 6. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $1 - \dfrac{9}{x^2} = \dfrac{3}{4}$ gives x^2 = 36, so x=6 is the only value in the domain; point (6; $\dfrac{15}{2}$). *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Projection and circle

1. $(1;1)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $\lambda = \dfrac{5+2-3}{2} = 2$; $\mathrm{H}(3;0)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $PH^2 = 8$; $(x-5)^2 + (y-2)^2 = 8$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV2-T2 · Mock exam 1

Practice paper . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14·16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 96 C. 100 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. 6 B. -6 C. $\frac{3}{2}$ D. $\frac{-3}{2}$

8. The slope between (0; 3) and (2; 9) is
   A. -3 B. 2 C. 6 D. 3

9. The mean of 3 , 5, 7 is
   A. 3 B. 5 C. 6 D. 15

10. A sector of 90 ° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. -1 D. 5

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Two draws (5 points)

An urn contains four red balls and two blue balls, indistinguishable to the touch. We draw two balls at random, one after the other, without replacing the first. At each draw, the balls still in the urn are equally likely. The third question uses a different protocol, with replacement.

1. Calculate the probability of getting two red balls. (1 pt)

2. Calculate the probability of getting one ball of each colour, taking into account the two possible orders. (2 pts)

3. The first ball is now put back before the second draw. Calculate the probability of getting one ball of each colour and compare it with that of the protocol without replacement. (2 pts)

Exercise 2 · One box (5 points)

We cut out squares with side x cm from the four corners of a square sheet with side 10 cm. Folding up the edges forms an open box. Its volume, in cubic centimetres, is $V(x) = x(10 - 2x)^2$. This model is studied on [0; 5], including the flat endpoint cases.

1. Calculate V' in factorised form. (1 pt)

2. Study the variation. (2 pts)

3. Determine the exact maximum volume and justify that the endpoints do not give the maximum. (2 pts)

Exercise 3 · Identification and intersection (4 points)

In an orthonormal coordinate system, consider the set $\mathcal{C}$ of the points $M(x; y)$ which satisfy $x^2 + y^2 + 4x - 2y - 4 = 0$. Once it is identified, we will study its intersections with a line and a tangent.

1. Show that $\mathcal{C}$ is a circle and give its centre and radius. (1 pt)

2. Determine all points common to $\mathcal{C}$ and to the line $x = -2$. (1 pt)

3. Write the equation of the tangent to the circle at the point of intersection with the larger y-coordinate. Justify its direction. (2 pts)

Solutions EV2-T2

MCQ: 1 D, 2 D, 3 A, 4 C, 5 C, 6 D, 7 D, 8 D, 9 B, 10 A, 11 D, 12 B. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Two draws

1. $(\frac{4}{6})(\frac{3}{5}) = \frac{2}{5}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $(\frac{4}{6})(\frac{2}{5}) + (\frac{2}{6})(\frac{4}{5}) = \frac{8}{15}$. *Marking: 2 points. Method or justification : 1 ; calculationation and conclusion : 1.*
3. With replacement: 2($\frac{2}{3})(\frac{1}{3}) = \frac{4}{9}$. The models differ; $\frac{8}{15} > \frac{4}{9}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · A box

1. $V\prime = 4(5-x)(5-3x)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Increases up to $\frac{5}{3}$ then decreases to 5. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $V(\frac{5}{3}) = \frac{2000}{27}$; zero values at the endpoints. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Identification and intersection

1. $(x+2)^2+(y-1)^2$ = 9: centre (-2;1), radius 3. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Points (-2; 4) and $(-2;-2)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Vertical radius at the upper point, horizontal tangent y = 4. Normal vector (0;1). *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV2-A · Mock exam 1

Paper reserved for the teacher . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14·16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $4 + 2 \times 3$.
   A. 13 B. 18 C. 10 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 100 C. 96 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. $\frac{4}{5}$ C. 1 D. $\frac{21}{20}$

7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -8 D. -2

8. The slope between (0; 4) and (2; 10) is
   A. -3 B. 3 C. 2 D. 6

9. The mean of 4, 6, 8 is
   A. 7 B. 18 C. 6 D. 4

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{1}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 16 B. 0 C. 6 D. 8

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · An unknown probability (5 points)

Two events A and B of the same random experiment satisfy $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{2}$ and $P_A(B) = \frac{3}{4}$. We want to complete the probabilities of a tree whose first level distinguishes A from its complement.

1. Calculate $P(A \cap B)$. (1 pt)

2. Determine $P_{\overline{A}}(B)$ and then complete a tree. (2 pts)

3. Determine $P_B(A)$ and discuss independence. (2 pts)

Exercise 2 · Relative position and variations (5 points)

The functions f and g are defined on $[-1; 4]$ by $f(x) = x^3 - 3x^2 + 2$ and $g(x) = 2$. Their curves are drawn in the same coordinate system. We must distinguish the variations of f from the position of its curve relative to that of g.

1. Calculate f' and study the variations. (1 pt)

2. Compare the graphs of f and g on the domain. (2 pts)

3. Determine the minimum and maximum of f on this domain. (2 pts)

Exercise 3 · A perpendicular bisector (4 points)

In an orthonormal coordinate system, we are given the points $A(0; 2)$, $B(6; 4)$ and $P(2; 6)$. Let I the midpoint of $[AB]$. We look for an equation of the perpendicular bisector of this segment, then use this line to compare two distances.

1. Find the midpoint of AB and a normal vector to its perpendicular bisector. (1 pt)

2. Determine the equation of the perpendicular bisector. (1 pt)

3. Is P equidistant from A and B? Check by two methods. (2 pts)

Solutions EV2-A

MCQ: 1 C, 2 D, 3 B, 4 C, 5 B, 6 C, 7 D, 8 B, 9 C, 10 D, 11 B, 12 D. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · An unknown probability

1. $\frac{3}{10}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $P(\overline{A} \cap B) = \frac{1}{2} - \frac{3}{10} = \frac{1}{5}$; divide by $\frac{3}{5}$ gives $\frac{1}{3}$; complements $\frac{1}{4}$ and $\frac{2}{3}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $P_B(A) = \frac{3}{5}$; different from $P(A) = \frac{2}{5}$, therefore not independent. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Relative position and variations

1. $f\prime = 3x(x-2)$; increases up to 0, decreases up to 2, then increases. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $f - g = x^2(x-3)$: negative for x<3 except for equality at 0, also zero at 3, positive after 3. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(-1) = -2, f(0) = 2, f(2) = -2$, f(4) = 18. Minimum -2 at -1 and 2; maximum 18 at 4. *Marking : 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · A perpendicular bisector

1. I(3;3), normal vector (6;2), or (3;1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $3x + y - 12 = 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. $3(2) + 6 - 12 = 0$, so yes; $PA^2 = 4 + 16 = 20,\ PB^2 = 16 + 4 = 20$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV2-B · Mock exam 1

Paper reserved for the teacher . Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14·16 January 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $5 + 2 \times 3$.
   A. 11 B. 10 C. 14 D. 21

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 10 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 40 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 100 B. 64 C. 96 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of $2x + 5 = 0$ is
   A. -10 B. $\frac{5}{2}$ C. $\frac{-5}{2}$ D. 10

8. The slope between (0; 5) and (2; 11) is
   A. 2 B. -3 C. 3 D. 6

9. The mean of 5, 7, 9 is
   A. 7 B. 21 C. 8 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 1 B. 7 C. 9 D. 20

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Independence and repetition (5 points)

Two events A and B are independent and satisfy $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. The first question concerns these events. In a separate experiment, used in questions 2 and 3, we repeat three times, independently, a trial whose probability of success is $\frac{1}{4}$.

1. Calculate $P(A \cap B)$ and $P(A \cup B)$. (1 pt)

2. Calculate the probability of exactly two successes in the three trials. (2 pts)

3. Calculate that of at least one success and justify using the complement. (2 pts)

Exercise 2 · Rational and symmetry (5 points)

The function f is defined on $\mathbb{R} \setminus \{0\}$ by $f(x) = \frac{x^2 + 4}{x}$. Its domain is symmetric about 0, but it is not an interval. We will use its parity to link the studies on the two half-lines.

1. Study parity and rewrite f. (1 pt)

2. Study the variation on]0; + ∞[. (2 pts)

3. Deduce the variation on] $-\infty$; 0[and explain why f is not increasing over its whole domain. (2 pts)

Exercise 3 · Distance and parameter (4 points)

In an orthonormal coordinate system, we are given the line d with equation $3x + 4y - 12 = 0$ and the point $P(0; 0)$. Let H be the orthogonal projection of P on d. We seek the circle with centre P tangent to this line.

1. Give a normal vector and check that (4; 0) is on d. (1 pt)

2. Calculate the projection H of P. (1 pt)

3. Determine the distance from P to d and the equation of the circle with centre P tangent to d. (2 pts)

Solutions EV2-B

MCQ: 1 A, 2 C, 3 D, 4 B, 5 A, 6 B, 7 C, 8 C, 9 A, 10 D, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Independence and repetition

1. $\frac{1}{6}$ and $\frac{1}{4}+\frac{2}{3}-\frac{1}{6}=\frac{3}{4}$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*

2. Three paths, each ($\frac{1}{4})^2(\frac{3}{4})$: total $\frac{9}{64}$. *Marking: 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

3. Complement of three failures: 1 $-(\frac{3}{4})^3=\frac{37}{64}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Rational and symmetry

1. $f=x+\frac{4}{x}$, odd. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. $f\prime=\frac{x^2-4}{x^2}$; decreases up to 2, then increases. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. Increases on $]-\infty;-2]$, decreases on $[-2;0[$. On [1; 2], f decreases: a counterexample to increasing everywhere. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Distance and Parameter

1. Normal vector (3;4); 12 $-12=0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. $\lambda=-\frac{12}{25}$, therefore $H(\frac{36}{25};\frac{48}{25})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

3. Distance $\frac{12}{5}$; $x^2+y^2=\frac{144}{25}$. *Marking: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV3-T1 · Mock exam 2

Practice paper . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1·3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $6 + 2 \times 3$.
   A. 12 B. 24 C. 11 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 96 C. 100 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$

7. The solution of $2x + 6 = 0$ is
   A. 12 B. -12 C. -3 D. 3

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 2 D. 6

9. The mean of 6 , 8, 10 is
   A. 9 B. 8 C. 24 D. 6

10. A sector of 90 ° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A compound gain (5 points)

A game consists of two independent trials, with a probability of success of $\frac{1}{2}$ on each trial. Each success earns 3 monetary units; taking part costs 2 units. Let X be the net gain, i.e. the total received minus the entry fee paid.

1. Determine the probability distribution of the net gain X. (1 pt)

2. Calculate its expectation and variance. (2 pts)

3. Determine the entry fee that would make the game fair. Explain what would happen to the standard deviation of the net gain. (2 pts)

Exercise 2 · Peak of a model (5 points)

A quantity is modelled for $t \geq 0$ by the function $f(t) = (t+2)e^{-t}$. The time t is expressed in a fixed unit. We seek its maximum value, then an approximation near the initial time.

1. Differentiate f. (1 pt)

2. Study the variation and give the maximum. (2 pts)

3. Write the tangent at 0 and approximate $f(0{,}1)$. (2 pts)

Exercise 3 · Unit circle (4 points)

The plane is equipped with an orthonormal coordinate system. On the unit circle with centre O, the points A and B are the respective images of the real numbers $\frac{\pi}{6}$ and $5\frac{\pi}{6}$.

1. Give their coordinates. (1 pt)

2. Calculate AB. (1 pt)

3. Give the equation of their perpendicular bisector and justify geometrically. (2 pts)

Solutions EV3-T1

MCQ: 1 A, 2 C, 3 D, 4 D, 5 C, 6 C, 7 C, 8 B, 9 B, 10 C, 11 C, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · A compound gain

1. Values -2,1,4; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*
2. $E(X) = 1$; $E(X^2) = 1 + \frac{1}{2} + 4 = \frac{11}{2}$; $V(X) = \frac{9}{2}$. *Marking: 2 points. Method or justification : 1 ; calculation and conclusion: 1.*
3. Mean gross gain 3: fair stake 3. Standard deviation unchanged $\frac{3}{\sqrt{2}}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Peak of a model

1. $f\prime = -(t+1)e^{-t}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Strictly decreasing on [0; + ∞[, maximum 2 at 0. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*
3. $y = 2 - t$; approximation 1.9. This is not an exact equality. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Unit circle

1. $A(\frac{\sqrt{3}}{2}; \frac{1}{2})$, $B(-\frac{\sqrt{3}}{2}; \frac{1}{2})$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*
2. $AB = \sqrt{3}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. Midpoint $(0; \frac{1}{2})$, AB horizontal; vertical perpendicular bisector $x = 0$. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV3-T2 · Mock exam 2

Practice paper . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1·3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 12 D. 8

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 60 C. 100 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of $2x + 2 = 0$ is
   A. -1 B. 4 C. -4 D. 1

8. The slope between (0; 2) and (2; 8) is
   A. 3 B. 2 C. 6 D. -3

9. The mean of $2, 4, 6$ is
   A. 2 B. 4 C. 5 D. 12

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Simulation and average (5 points)

The random variable X takes the values 0, 2 and 6 with the respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. We then simulate samples of 25 independent observations of this variable; let m be the mean observed in a sample.

1. Calculate the expectation $E(X)$. (1 pt)

2. Calculate $V(X)$ and then the exact standard deviation $\sigma(X)$. (2 pts)

3. Write the interval of the values of m satisfying $|m - E(X)| \leq \frac{2\sigma(X)}{5}$. Explain why a simulation over a finite number of samples does not guarantee an exact coverage percentage. (2 pts)

Exercise 2 · Compare two models (5 points)

We define on $\mathbb{R}$ the functions $f(x) = e^{2x}$ and $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of f at whole-number times, from index 0.

1. Calculate the derivative of $h = f - g$. (1 pt)

2. Study h and compare the curves. (2 pts)

3. For $u_n = e^{2n}$, give the common ratio and the sum of the first four terms. (2 pts)

Exercise 3 · A translated circle (4 points)

In an orthonormal coordinate system, consider the circle $\mathcal{C}$ with equation $(x-1)^2 + (y+1)^2 = 4$ and the line d with equation $y = -1$. Let A and B be their intersection points, A being the one with the smaller x-coordinate.

1. Give centre and radius. (1 pt)

2. Determine the intersections with the line. (1 pt)

3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. (2 pts)

Solutions EV3-T2

MCQ: 1 D, 2 D, 3 B, 4 D, 5 C, 6 C, 7 A, 8 A, 9 B, 10 A, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Simulation and average

1. $E(X) = 0 + \frac{1}{2} + \frac{3}{2} = 2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $E(X^2)$ = 1 + 9 = 10, variance 6, standard deviation $\sqrt{6}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $[2 - \frac{2\sqrt{6}}{5}; 2 + \frac{2\sqrt{6}}{5}]$. A frequency over a finite number of simulations fluctuates. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Compare two models

1. $h\prime = 2(e^{2x} - 1)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Decreases down to 0 then increases; minimum 0. Hence $f \geq g$ with equality only at 0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Common ratio e^2 ; sum 1 + $e^2 + e^4 + e^6 = \frac{1 - e^8}{1 - e^2}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · A translated circle

1. Centre (1;-1), radius 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. A(-1;-1), B(3;-1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. AB is a diameter, so the product is 0, endpoints included; identity $MI^2 - 4 = 0$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV3-A · Mock exam 2

Paper reserved for the teacher . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1·3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 15 C. 8 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{5}{3}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 1 C. 8 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 60 C. 100 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. -6 D. 6

8. The slope between (0; 3) and (2; 9) is
   A. 6 B. -3 C. 2 D. 3

9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. 5 D. -1

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Lottery and dispersion (5 points)

A lottery pays gross winnings G of 0, 4 or 12 monetary units with respective probabilities $\frac{1}{2}$, $\frac{3}{8}$ and $\frac{1}{8}$. To take part, you pay 3 units. Let $X = G - 3$ be the net gain.

1. Give the distribution of the net gain X. (1 pt)

2. Calculate the expectation and variance of X. (2 pts)

3. The gross winnings and the stake are doubled. What happens to the mean and standard deviation of the net gain? (2 pts)

Exercise 2 · Derivative and horizontal tangent (5 points)

The function f is defined on $\mathbb{R}$ by $f(x) = (2x - 3)e^x$. We study its extremum and the tangent to its curve at the point with x-coordinate 0. All values are expected in exact form.

1. Calculate $f(0)$, then $f'(x)$ and $f'(0)$. (1 pt)

2. Study the variation of f on $\mathbb{R}$ and give its exact extremum. (2 pts)

3. Determine the equation of the tangent at the point with x-coordinate 0. State whether its point of contact is above or below the x-axis. (2 pts)

Exercise 3 · Angle and length (4 points)

In an orthonormal coordinate system, we are given the points $A(0;0)$, $B(4;0)$ and $C(2;2\sqrt{3})$. We want to determine the type of the triangle ABC by comparing lengths and the scalar product.

1. Calculate AB and AC. (1 pt)

2. Determine the cosine, then the angle BAC in radians. (1 pt)

3. Determine BC and the type of triangle. (2 pts)

Solutions EV3-A

MCQ: 1 A, 2 B, 3 A, 4 C, 5 C, 6 D, 7 B, 8 D, 9 D, 10 B, 11 A, 12 B. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Lottery and dispersion

1. X takes the values -3,1,9 with the same probabilities. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. $E(X) = -\frac{3}{2} + \frac{3}{8} + \frac{9}{8} = 0$. $E(X^2) = \frac{9}{2} + \frac{3}{8} + \frac{81}{8}$ = 15; variance 15. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. Net gain 2X, expectation 0; standard deviation $2\sqrt{15}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Derivative and horizontal tangent

1. $f(0) = -3$; $f\prime = (2x-1)e^x$, $f\prime(0) = -1$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. Decreases down to $\frac{1}{2}$, then increases; minimum $-2e^{\frac{1}{2}}$. *Marking: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $y = -x - 3$; point of contact (0;-3), below the horizontal axis. *Marking: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Angle and length

1. AB=4, $AC = \sqrt{4+12} = 4$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Product 8; cosine $\frac{8}{16} = \frac{1}{2}$, angle $\frac{\pi}{3}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

3. $BC = \sqrt{4+12}$ = 4: equilateral triangle. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV3-B · Mock exam 2

Paper reserved for the teacher . Content of Chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1·3 April 2027.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $4 + 2 \times 3$.
   A. 10 B. 9 C. 18 D. 13

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{5}{3}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 4 B. 1 C. 2 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 60 C. 10 D. 40

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 96 B. 60 C. 64 D. 100

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 2 B. -3 C. 3 D. 6

9. The mean of $4, 6, 8$ is
   A. 7 B. 18 C. 6 D. 4

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{1}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 16 B. 0 C. 6 D. 8

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Four trials and a gain (5 points)

A game consists of four independent trials; each trial succeeds with probability $\frac{1}{2}$. The gross gain G is 8 monetary units if all four trials succeed, and 0 otherwise. A stake may be set to make the game fair.

1. Calculate the probability of receiving 8. (1 pt)

2. Calculate the expectation and variance of G. (2 pts)

3. Determine the fair stake; explain whether the expectation is a possible gain. (2 pts)

Exercise 2 · A derivative of constant sign (5 points)

Consider the function f defined on $\mathbb{R}$ by $f(x) = (x^2 + 1)e^x$. We want to check whether a horizontal tangent necessarily corresponds to an extremum.

1. Calculate and factorise f'. (1 pt)

2. Study the variation; does the horizontal tangent at -1 give an extremum? (2 pts)

3. Write the equation of the tangent at 0, then calculate $f(1)$ exactly. (2 pts)

Exercise 3 · Projection onto a diagonal (4 points)

In an orthonormal coordinate system with origin O, we consider $P(2; 0)$ and the line d with equation $y = x$. Let H be the orthogonal projection of P on d. The unit vector requested at the end must point from O to H.

1. Give a non-zero normal vector to the line d. (1 pt)

2. Determine the coordinates of H, then the distance PH. (1 pt)

3. Give the unsigned angle $\widehat{POH}$ in radians, then the coordinates of the vector $\dfrac{\overrightarrow{OH}}{OH}$. (2 pts)

Solutions EV3-B

MCQ: 1 A, 2 B, 3 A, 4 B, 5 D, 6 B, 7 C, 8 C, 9 C, 10 D, 11 D, 12 D. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Four trials and a gain

1. $(\frac{1}{2})^4 = \frac{1}{16}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $E(G) = \frac{1}{2}$, $E(G^2) = 4$; $V(G) = \frac{15}{4}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Stake $\frac{1}{2}$. The expected gross gain is $\frac{1}{2}$, which is not a possible result of a single game. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · A derivative of constant sign

1. $f\prime = (x^2+2x+1)e^x = (x+1)^2e^x$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*
2. Positive except at -1; f is strictly increasing (on both sides and across the join). No extremum at -1. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(0) = 1, f\prime(0) = 1 : y = x+1$; $f(1) = 2e$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Projection onto a diagonal

1. Normal vector (1;-1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. H(1;1), PH= $\sqrt{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. Angle $\frac{\pi}{4}$; vector ($\frac{\sqrt{2}}{2}; \frac{\sqrt{2}}{2}$). *Marking: 2 points. Method or justification: 1; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV4-T1 · EAM preparation

Practice paper . The whole syllabus.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 21 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 60 B. 96 C. 100 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$

7. The solution of $2x + 5 = 0$ is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. -10

8. The slope between (0; 5) and (2; 11) is
   A. 3 B. -3 C. 6 D. 2

9. The mean of 5 , 7, 9 is
   A. 7 B. 21 C. 5 D. 8

10. A sector of 90 ° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 20 B. 7 C. 1 D. 9

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Choosing a distribution (5 points)

We compare two games. The first gives a gain $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and a gain $Y = 8$ with probability $\frac{1}{3}$. Gains are expressed in the same currency unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(Y)$ and compare the spreads of the two gains. (2 pts)

3. We play both games and add the winnings. Calculate $E(X+Y)$, citing the property used. (2 pts)

Exercise 2 · Discrete threshold and continuous model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at integer times $n \in \mathbb{N}$ and we set $u_n = Q(n)$. For comparisons without a calculator, use the bounds $e^2 < 8 < e^3$.

1. Study the direction of variation. (1 pt)

2. Determine the first integer n such that $Q(n) < 8$ using the given bounds. (2 pts)

3. Give the type of $u_n = Q(n)$ and the expression of $u_0 + \cdots + u_3$. (2 pts)

Exercise 3 · Equation and circle (4 points)

In an orthonormal coordinate system, consider the set $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. The aim is to determine its characteristics, then to identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and state its centre and radius. (1 pt)

2. Determine its intersection points with the line with equation $x = 1$. (1 pt)

3. Determine the equation of the tangent to the circle at its highest point and justify. (2 pts)

Solutions EV4-T1

MCQ: 1 B, 2 B, 3 B, 4 C, 5 C, 6 B, 7 C, 8 A, 9 A, 10 A, 11 A, 12 D. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Choosing a distribution

1. $E(X) = 2$; $E(Y) = -\frac{2}{3} + \frac{8}{3} = 2$. *Marking: 1 point. Result and basic justification expected ; half a point possible for a correct but unfinished method.*
2. $E(Y^2) = \frac{2}{3} + \frac{64}{3}$ = 22, variance 18, versus 0 for X. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*
3. Linearity: expectation 4. Moreover, X is constant. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Discrete threshold and continuous model

1. $Q\prime = -64e^{-t} < 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $Q(2) > 8$, $Q(3) < 8$; decreasing: first index 3. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Geometric, first term 64, common ratio e^{-1} ; sum $\frac{64(1-e^{-4})}{1-e^{-1}}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Equation and circle

1. $(x-1)^2+(y-2)^2$ = 4; centre (1;2), radius 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. (1;0) and (1;4). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. Vertical radius at point (1;4), horizontal tangent $y = 4$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV4-T2 · EAM preparation

Practice paper . The whole syllabus.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $6 + 2 \times 3$.
   A. 24 B. 15 C. 11 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 64 B. 100 C. 96 D. 60

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of $2x + 6 = 0$ is
   A. -3 B. 12 C. 3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. 2 B. -3 C. 3 D. 6

9. The mean of 6 , 8, 10 is
   A. 6 B. 24 C. 8 D. 9

10. A sector of 90 ° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · A reversed tree (5 points)

Two events A and B satisfy $P(A) = \frac{3}{5}$, $P_A(B) = \frac{1}{3}$ and $P_{\overline{A}}(B) = \frac{3}{4}$. We want to construct a tree that starts by distinguishing B and $\overline{B}$, instead of A and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)

2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. (2 pts)

3. Construct and complete the weighted tree whose first level distinguishes B and $\overline{B}$, then the second A and $\overline{A}$. (2 pts)

Exercise 2 · Two optimisation methods (5 points)

The function f is defined on the interval [1; 6] by $f(x) = x(8 - x)$. We seek its extrema on this specific domain, then the values of x for which $f(x)$ reaches at least 15.

1. Give a vertex form and a maximum. (1 pt)

2. Find the maximum again by differentiation. (2 pts)

3. Determine the minimum on the domain and solve $f(x) \geq 15$. (2 pts)

Exercise 3 · Arguments and symmetry (4 points)

In an orthonormal coordinate system with origin O, the points A and B on the unit circle are the images of $\frac{7\pi}{6}$ and $11\frac{\pi}{6}$. The unsigned angle $\widehat{AOB}$ will be given in radians, between 0 and π.

1. Give their exact coordinates. (1 pt)

2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)

3. Determine the unsigned angle AOB, then AB. (2 pts)

Solutions EV4-T2

MCQ: 1 D, 2 A, 3 B, 4 A, 5 B, 6 C, 7 A, 8 C, 9 C, 10 D, 11 D, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · A reversed tree

1. $P(A \cap B) = \frac{1}{5}$; $P(B) = \frac{1}{5} + \frac{3}{10} = \frac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $P_B(A) = \frac{2}{5}$; $P(A \cap \overline{B}) = \frac{2}{5}$ so $P_{\overline{B}}(A) = \frac{4}{5}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. First level $\frac{1}{2}, \frac{1}{2}$; after B: $\frac{2}{5}$ and $\frac{3}{5}$; after the complement of B: $\frac{4}{5}$ and $\frac{1}{5}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Two optimisation methods

1. $f = 16 - (x-4)^2$; maximum 16 at 4. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $f\prime = 8 - 2x$, positive before 4 and negative after. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Endpoints: f(1)=7, f(6)=12; minimum 7. Inequality ($x-4)^2 \leq 1 : [3;5]$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Arguments and symmetry

1. $A(-\frac{\sqrt{3}}{2}; -\frac{1}{2}),\ B(\frac{\sqrt{3}}{2}; -\frac{1}{2})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $-\frac{3}{4} + \frac{1}{4} = -\frac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
3. Angle $\frac{2\pi}{3}$; $AB = \sqrt{3}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV4-A · EAM preparation

Paper reserved for the teacher . The whole programme.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 8 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 1 C. 7 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 100 B. 60 C. 96 D. 64

6. Increase of 25% then decrease of 20%: overall multiplier?
   A. 1 B. $\frac{4}{5}$ C. $\frac{21}{20}$ D. $\frac{9}{10}$

7. The solution of $2x + 2 = 0$ is
   A. 1 B. 4 C. -1 D. -4

8. The slope between (0; 2) and (2; 8) is
   A. -3 B. 3 C. 6 D. 2

9. The mean of 2, 4, 6 is
   A. 4 B. 5 C. 2 D. 12

10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · Two related variables (5 points)

The random variable X takes the values 1, 2 and 3, each with probability $\frac{1}{3}$. On the same experiment, define the variable $Y = 4 - X$. We want to check whether a rule valid for expectation remains true for variance.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(X)$, $V(Y)$ and $V(X+Y)$. (2 pts)

3. Use these results to disprove the statement: "The variance of a sum is always equal to the sum of the variances." (2 pts)

Exercise 2 · Comparing a curve and a line (5 points)

The functions $f(x) = e^{-x}$ and $g(x) = 1 - x$ are defined on $\mathbb{R}$. Set $h = f - g$. The comparison of their curves will then make it possible to check the direction of an approximation.

1. Differentiate their difference. (1 pt)

2. Prove $f(x) \geq g(x)$ for every real x. (2 pts)

3. Write the affine approximation of $e^{-0{,}02}$ and state whether it is an underestimate or an overestimate. (2 pts)

Exercise 3 · Normal vector and radius (4 points)

In an orthonormal coordinate system, consider the point $I(1; 1)$ and the line d with equation $3x + 4y - 17 = 0$. Let H be the orthogonal projection of I on d. We seek the circle with centre I tangent to d.

1. Calculate the distance from I to d. (1 pt)

2. Calculate the projection H. (1 pt)

3. Give the equation of the circle with centre I tangent to d and check that H lies on the circle. (2 pts)

Solutions EV4-A

MCQ: 1 C, 2 C, 3 D, 4 C, 5 A, 6 A, 7 C, 8 B, 9 A, 10 A, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · Two related variables

1. Both expectations are equal to 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. $E(X^2) = \frac{14}{3}$, therefore $V(X) = \frac{2}{3}$; same variance for Y. X+Y=4, variance 0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $0 \neq \frac{2}{3} + \frac{2}{3}$. The counterexample uses one sample space; Y depends on X. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · Comparing a curve and a line

1. $h\prime = 1 - e^{-x}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. Negative if x<0, positive if x>0; minimum h(0)=0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. Approximation 0.98; an underestimate, by the inequality. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Normal Vector and Radius

1. $\frac{|3 + 4 - 17|}{5} = 2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*

2. $\lambda = -\frac{10}{25} = -\frac{2}{5}$; H$(\frac{11}{5}; \frac{13}{5})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

3. $(x-1)^2 + (y-1)^2$ = 4; for H, ($\frac{6}{5})^2 + (\frac{8}{5})^2 = 4$. *Marking: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.

EV4-B · EAM preparation

Paper reserved for the teacher . The whole programme.

Duration: 2 h. No calculator. Marked out of 20. May and June practice; regional EAM to be confirmed.

Part 1 · Automatic skills (6 points)

Twelve questions, 0.5 point each. A single correct answer; no justification required. An incorrect or missing answer scores 0, with no points deducted.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 12 C. 15 D. 8
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 8 B. 2 C. 4 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10
5. A value of 80 is obtained after a decrease of 20%. Initial value?
   A. 60 B. 64 C. 96 D. 100
6. Increase of 25% then decrease of 20%: overall multiplier?
   A. $\frac{21}{20}$ B. 1 C. $\frac{4}{5}$ D. $\frac{9}{10}$
7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. 6 C. $\frac{-3}{2}$ D. -6
8. The slope between (0; 3) and (2; 9) is
   A. 6 B. -3 C. 3 D. 2
9. The mean of $3, 5, 7$ is
   A. 3 B. 15 C. 6 D. 5
10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability equals $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 5 C. -1 D. 7

Part 2 · Independent exercises (14 points)

Every step must be justified. Exact answers are expected unless otherwise stated.

Exercise 1 · An average price (5 points)

A service offers several price plans. 60 per cent of customers choose plan A; among them, half pay 0 units and the other half pay 4 units. All other customers pay 6 units. We select a customer at random and let X be the price they pay. The last question studies a uniform discount, including for the price that was initially zero.

1. Determine the probability distribution of X. (1 pt)

2. Calculate its expectation and variance. (2 pts)

3. All prices decrease by 1 unit. Calculate the new mean and the new standard deviation. Interpret the negative price obtained for some customers. (2 pts)

Exercise 2 · exponential maximum (5 points)

The function f is defined on [0; 4] by $f(t) = t^2 e^{-t}$. We seek its maximum on this interval. For the last comparison without a calculator, you may use the given inequality $e^2 > 4$.

1. Calculate f' in factorised form. (1 pt)

2. Determine the variation and the maximum. (2 pts)

3. Compare exactly $f(4)$ with the maximum, given $e^2 > 4$. (2 pts)

Exercise 3 · Determine a position (4 points)

In an orthonormal coordinate system, the points $A(0; 0)$ and $B(4; 0)$ are fixed. The point M moves on the line with equation $y = 2$; its coordinates are therefore $M(x; 2)$, where x ranges over $\mathbb{R}$.

1. Calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$. (1 pt)

2. Determine the values of x for which the two vectors are orthogonal. (1 pt)

3. Interpret the result using the circle with diameter AB. (2 pts)

Solutions EV4-B

MCQ: 1 A, 2 B, 3 C, 4 C, 5 D, 6 B, 7 C, 8 C, 9 D, 10 C, 11 A, 12 D. Each correct answer: 0.5 point; total 6.

Fluency checks. Order of operations; sum of fractions; subtraction of exponents; multiplier 1.2; division by 0.8; product 1.25 times 0.8; solving a linear equation; vertical change divided by horizontal change; mean; angle divided by 360; complement to 1; substitution inside brackets.

Exercise 1 · An average price

1. Values 0,4,6; probabilities 0.3; 0.3; 0.4. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. $E(X) = 3{,}6 = \frac{18}{5}$; $E(X^2) = 19{,}2 = \frac{96}{5}$; variance $\frac{156}{25}$. *Marking: 2 points. Method or justification : 1; calculation and conclusion: 1.*
3. Mean $\frac{13}{5}$; standard deviation $\frac{2\sqrt{39}}{5}$, unchanged. A price of -1 would be a refund: check whether this policy is actually planned. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 2 · exponential maximum

1. $f\prime = t(2-t)e^{-t}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished method.*
2. Increases from 0 to 2 then decreases; maximum $4e^{-2}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(4) = 16e^{-4}$; ratio to maximum 4 $e^{-2} < 1$ since $e^2 > 4$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Exercise 3 · Determine a position

1. $(-x;-2)\cdot(4-x;-2) = x^2 - 4x + 4 = (x-2)^2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Zero product only at x=2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Circle with centre (2;0), radius 2; the line y=2 is tangent to it at (2;2). *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of checking. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error carried forward consistently. The proposed mark scheme does not replace a moderation decision by the team.